SUPPLEMENT TO "THE GENERALIZED LAX CONJECTURE HOLDS FOR MIXED DETERMINANTS"

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ABSTRACT. This supplement records two groups of results outside the main line of the paper "The generalized Lax conjecture holds for mixed determinants": the scalar-specialization bounds obtained from the kernel of the completely positive construction, which the cubic kernel of the paper supersedes but which keep the maps completely positive over $\mathbb{C}$, and the order-244 sharpness of constant compression for one explicit positive definite matrix, together with a different pencil of order at most 20 for the same matrix. Numbers without the prefix S refer to the paper, whose notation is used throughout.

S1. SCALAR BOUNDS FROM THE COMPLETELY POSITIVE KERNEL

The paper bounds the orders of fixed-matrix pencils with the boundary kernel continued from the cubic pencil of Proposition 4.3. In this section we record the corresponding bounds for the kernel v_d of the completely positive construction (Appendix A.1), with the degrees $\kappa_{d,m}$ and $\delta_{d,m}$ of (4.1). The bounds are larger, but the maps stay completely positive over $\mathbb{C}$; they fill the CP columns of the table in Section 5.1. For $m \geq 2$, every coordinate of v_d acquires a common monomial factor under scalar specialization, which lowers the degree that the coefficient count sees. We define

$$\alpha_{2,m} = 0, \qquad \alpha_{d+1,m} = 2\kappa_{d,m} - 1 + (m-1)\alpha_{d,m},$$
$$E_{d,m} = \delta_{d,m} - m\alpha_{d,m}.$$

Every monomial occurring in v_d has degree at least $\alpha_{d,m}$ in the entries of each individual argument. To check the induction, we scale the full bordered argument c by t and work at $t = 0$. On a nonempty generic chart, D_c has valuation two and Y_c remains bounded, so w_c and $z_c = U_c^* w_c / D_c$ have valuations at least $2\kappa_{d,m}$ and $2\kappa_{d,m} - 1$. For $j \neq c$, $D_j(0) \neq 0$ and the cth argument of Y_j is still tA_c; hence both w_j and z_j have valuation at least $\alpha_{d,m}$. The tensor formulas give the stated recurrence. The degree bounds are polynomial identities, so they extend off the chart. In particular, $v_d(z_1 B_1, \ldots, z_m B_m)$ is divisible by $\prod_j z_j^{\alpha_{d,m}}$.

Proposition S1.1 (Size after scalar specialization)**.** *Let $d, m \geq 2$, let $B_1, \ldots, B_m \succeq 0$, and suppose that $p_{d,m}(B_1, \ldots, B_m) > 0$. Put $f_B(z) = p_{d,m}(z_1 B_1, \ldots, z_m B_m)$, and use the count $H_{m,d}$ from (5.2). There is a constant isometry Q, allowed to depend on B, such that $M_B(z) = \sum_{j=1}^m z_j Q^* \Phi_j(B_j) Q$ has order at most $\min\{N_{d,m}, H_{m,d}(E_{d,m})\}$, represents the open and closed hyperbolicity cones of f_B with direction $\mathbf{1}$, and has determinant divisible by f_B. Its coefficients are positive semidefinite, and the maps $X \mapsto Q^* \Phi_j(X) Q$ remain unital and completely positive. For real symmetric B_j, Q can be real; for Hermitian B_j, it can be complex.*

Proof. We write $p = p_{d,m}$, $\delta = \delta_{d,m}$, $\alpha = \alpha_{d,m}$, and $E = E_{d,m}$. We treat the entries of $B_1, \ldots, B_m$ as independent variables and put $c(z) = \prod_j z_j^{\alpha}$. The preceding divisibility gives us polynomial vectors

$$\widetilde{v}(B, z) = c(z)^{-1} v_d(z_1 B_1, \ldots, z_m B_m), \qquad L(z_1 B_1, \ldots, z_m B_m)\widetilde{v} = f_B \widetilde{g}.$$

Here $\widetilde{g}$ is polynomial as well: with B still indeterminate, f_B has no z_j factor when $m \geq 2$, so the original kernel identity implies that c divides the specialized g_d. The vector $\widetilde{v}$ has degree E in z and degree δ in B.

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We first justify a generic nonvanishing condition within this family of sections. Proposition 3.1 and its proof remain valid over $\mathbb{C}$. We choose a coordinate h of v_d that is nonzero on a real symmetric boundary chart, so $p \nmid h$, and let $\widetilde{h} = c^{-1}h(z_1B_1, \dots, z_mB_m)$. After inverting all the z_j, the change of variables $X_j = z_jB_j$ identifies f_B with the irreducible polynomial $p(X)$. It also shows that f_B does not divide $\widetilde{h}$. Before localization, any additional factor of f_B would have to be a monomial in the z_j, which is impossible because p remains nonzero when any one argument is zero. Thus f_B is irreducible, and is coprime to $\widetilde{h}$, also over the fraction field $\mathbb{C}(B)$ by Gauss's lemma.

Hence the resultant

$$R(B, u) = \mathrm{Res}_t(f_B(u + t\mathbf{1}), \widetilde{h}(B, u + t\mathbf{1}))$$

is not identically zero. The same argument applies to real symmetric indeterminates. We may therefore choose u and real symmetric positive definite matrices $C_1, \dots, C_m$, all with rational coordinates, such that $R(C, u) \neq 0$: these tuples form a nonempty real open set. The resultant is homogeneous in B, since its two inputs have respective B-degrees d and δ and generic t-degrees d and E. Consequently $R(B + \varepsilon C, u)$ is a nonzero polynomial in ε, with nonzero leading coefficient $R(C, u)$. It is nonzero for all sufficiently small positive ε.

We temporarily write B for one of these perturbed tuples. Then f_B and $\widetilde{h}(B, \cdot)$ are coprime in z. Indeed, any nonconstant homogeneous common factor a satisfies $a(\mathbf{1}) \neq 0$, because $f_B(\mathbf{1}) > 0$. Thus $a(u + t\mathbf{1})$ has degree $\deg a > 0$, contradicting the nonzero line resultant. This argument does not require f_B to be irreducible. We divide each coordinate of $\widetilde{v}$ by f_B as a polynomial in the coordinate along $\mathbf{1}$. The leading coefficient $f_B(\mathbf{1}) > 0$ is constant in the remaining coordinates. The resulting remainder r is homogeneous of degree E and has degree less than d in that coordinate. It has at most $H_{m,d}(E)$ monomials, and satisfies $L(z_1B_1, \dots, z_mB_m)r = f_B\gamma$ for a polynomial vector γ. Its coordinate obtained from $\widetilde{h}$ remains coprime to f_B.

Let Q be an orthonormal basis for the coefficient span of r. Writing $r = Qw$ and multiplying the compressed kernel identity by the adjugate of M_B shows that f_B divides $(\det M_B)w_i$ for every i. The coprime coordinate of r is a constant linear combination of the w_i, so $f_B \mid \det M_B$. The positive definite locus of M_B is convex, contains $\mathbf{1}$, and avoids $\{f_B = 0\}$. Compression supplies the reverse inclusion from the open cone of f_B; taking closures gives the closed-cone equality.

We now return to the original tuple and let ε tend to zero, passing to a subsequence of the isometries with constant order and limit Q. Strict feasibility persists because $L(B_1, \dots, B_m) \succ 0$. The determinant factorization persists too: if $\det M_\varepsilon = f_\varepsilon q_\varepsilon$, multiplication by the nonzero limit polynomial f_B is an injective linear map between fixed finite-dimensional coefficient spaces. Its smallest singular value is positive and stays bounded below for nearby polynomials, so the coefficients of q_ε are bounded and a subsequence converges. Thus $f_B \mid \det M_B$, and the same positive-component argument proves both cone equalities at the limit. Compression preserves the asserted positivity and unital complete positivity. □

Proposition S1.2 (One compression for all common matrices)**.** *Let $d, m \geq 2$, and set $\delta = \delta_{d,m}$, $\alpha = \alpha_{d,m}$, and $E = \delta - m\alpha$. There are unital completely positive maps $\Psi_1, \dots, \Psi_m$, each with fixed real algebraic Kraus operators, such that, for every $A \succeq 0$ with $\eta(A, \dots, A) > 0$, the pencil $\sum_j z_j\Psi_j(A)$ represents the open and closed cones of $f_A(z) = \eta(z_1A, \dots, z_mA)$ and its determinant is divisible by f_A. Their common matrix order is at most the smaller of $\widehat{N}_{d,m}$ and*

$$\binom{m + E - 1}{m - 1}\binom{\ell + \delta}{\ell}, \qquad \ell = d^2 - 1. \tag{S1.1}$$

The same maps work over $\mathbb{R}$ and $\mathbb{C}$. For real symmetric A alone one may replace ℓ by $d(d+1)/2 - 1$. At $d = m = n$, the bound is $\exp(O(n^3 \log n))$.

Proof. We choose real symmetric positive definite matrices with rational entries $B_1, \dots, B_m$ for which the generic nonvanishing argument in Proposition S1.1 applies, and let h be the chosen coordinate

of v_d and $\widetilde{h}$ its divided restriction. At this tuple, f_B has no coordinate factor z_j: setting $z_j = 0$ leaves the nonzero term $\det(B_k)z_k^d$ for any $k \neq j$. Hence coprimality with the divided coordinate $\widetilde{h}$ also gives coprimality with the original restricted coordinate $h(z_1B_1, \dots, z_mB_m) = (\prod_j z_j^\alpha)\widetilde{h}(B, z)$. For $\varepsilon > 0$ we put

$$F_{A,\varepsilon}(z) = (z_1(A + \varepsilon B_1), \dots, z_m(A + \varepsilon B_m)), \qquad r_{A,\varepsilon}(z) = \frac{v_d(F_{A,\varepsilon}(z))}{\prod_j z_j^\alpha}.$$

The color-degree bound makes r polynomial. It has degree E in z and degree δ jointly in the entries of A and ε. Its coefficient of $A^\mu z^\beta$ is therefore $\varepsilon^{\delta-|\mu|}$ times a fixed vector. The span W of these coefficient vectors is independent of $\varepsilon \neq 0$. Since v_d omits the last diagonal sum, A_{dd} does not occur. Counting the degree-E monomials in z and the monomials of degree at most δ in the remaining ℓ entries of A gives (S1.1). In the Hermitian case we count formal ordered off-diagonal entries, as in the universal compression argument. The coefficient vectors are rational and real, so W has a real algebraic orthonormal basis Q.

We set $\Psi_j(X) = Q^*\Phi_j(X)Q$. For each fixed $\varepsilon > 0$, the restrictions of $p_{d,m}$ and h along $F_{A,\varepsilon}$ are coprime for generic A: a line resultant witnessing coprimality is nonzero at $A = 0$. Dividing the restricted kernel by $\prod_j z_j^\alpha$ preserves coprimality and its polynomial kernel identity. The coefficient compression argument therefore gives the determinant divisibility and cone equalities for generic positive definite A. They extend to all $A \succeq 0$ by the limiting argument in Proposition S1.1, since $L_{d,m}(F_{A,\varepsilon}(\mathbf{1})) \succeq \varepsilon(\sum_j \lambda_{\min}(B_j))I$. Finally we let $\varepsilon \downarrow 0$. We use the same Q throughout, and $\eta(A, \dots, A) > 0$ ensures strict feasibility at $\mathbf{1}$ in the limit. Divisibility and the cone equalities again pass to the limit. Compression preserves unital complete positivity. If the universal completely positive compression is smaller, we retain it instead. The asymptotic bound follows from $E \leq \delta \leq (d-1)(2m)^{d-2}$. □

The maps in Proposition S1.2 are independent of A. Allowing the compression to depend on A gives the smaller bound in Proposition S1.1; neither bound applies to the full universal cone $\mathcal{K}_{d,m}$.

For $d = m = n$ and every fixed PSD matrix A with positive diagonal, Proposition S1.1 gives $N_{\text{fixed}}(A) \leq H_{n,n}(E_{n,n})$, and hence $\log N_{\text{fixed}}(A) = O(n^2 \log n)$. Indeed, $E_{n,n} \leq (n-1)(2n)^{n-2}$, and the first binomial in $H_{n,n}$ already has this logarithmic bound. The bounds of this section use the kernel of the completely positive construction and its degrees $\kappa_{d,m}, \delta_{d,m}, \alpha_{d,m}, E_{d,m}$.

S2. Sharpness of constant compression, and a smaller pencil

Proposition 5.5 bounds the order of a fixed-matrix pencil at $d = m = n = 4$ by 244. In this section we show that for one explicit positive definite matrix no constant compression of the uncompressed pencil does better, and that a different construction nevertheless represents the same cone in order at most 20. A constant compression of $L(z)$ is a pencil $Q^*L(z)Q$, where Q is fixed and has full column rank. For the following matrix, the coefficient span is the smallest subspace that preserves the cone under such a compression.

Proposition S2.1. *Let*

$$A = \begin{pmatrix} 7 & 1 & 2 & 1 \\ 1 & 8 & 1 & 2 \\ 2 & 1 & 9 & 3 \\ 1 & 2 & 3 & 10 \end{pmatrix}, \qquad f(z) = \eta(z_1A, z_2A, z_3A, z_4A). \tag{S2.1}$$

Form the uncompressed pencil of order 1 639 792 *at* $(d, m) = (4, 4)$ *by one bordering step from Proposition 4.3, using the maps in its proof, and specialize its arguments to* z_jA. *The smallest order of a cone-preserving constant compression of this pencil is* 244.

Proof. The leading principal minors of A are $7, 55, 460, 3976$, so $A \succ 0$. The partition formula gives

$$f = 3976m_{(4)} + 17800m_{(3,1)} + 27676m_{(2,2)} + 57864m_{(2,1,1)} + 120960m_{(1,1,1,1)}, \tag{S2.2}$$

where m_λ denotes a monomial symmetric polynomial in four variables. In particular $f(\mathbf{1}) = 1\,210\,888 > 0$.

The quartic kernel obtained from the cubic kernel (4.8) by one step of the induction has degree 15 and is divisible by $c(z) = z_1z_2z_3z_4$. We divide by c and take the remainder modulo f in the coordinate t defined by $z = (x_1+t, x_2+t, x_3+t, t)$. The resulting vector R is homogeneous of degree 11 and has t-degree less than four, so its coefficient span has dimension at most $\binom{14}{3} - \binom{10}{3} = 244$.

We prove equality by an exact certificate computed in the finite field $\mathbb{F}_{1009}$. It uses 280 points of $\{f = 0\}$ with nonzero coordinates and exit denominators, evaluates the uncompressed kernel vectors there, computes their Gram matrix by multiplying inner products of their tensor factors, and exhibits a 244×244 minor with determinant 298 modulo 1009. The nonzero minor certifies the rank in characteristic zero for the following reason. The coefficients of R lie in $\mathbb{Z}_{(1009)}$: the original kernel is integral, division by c is polynomial, and the remainder operation only divides by powers of $f(\mathbf{1}) \equiv 88 \pmod{1009}$. At each recorded point the uncompressed kernel equals cR, so its value lies in the reduced coefficient span of R. The rank of a Gram matrix is at most the rank of its evaluated vectors, which is at most the rank of that coefficient matrix. The nonzero minor thus proves rank at least 244 modulo 1009, and hence over $\mathbb{Q}$.

The same computation gives the Gröbner basis [1] for the four partial derivatives of (S2.2) on the chart $z_4 = 1$ over $\mathbb{F}_{1009}$. Symmetry covers the other projective charts. Thus the projective quartic is smooth modulo 1009, hence smooth in characteristic zero: the common zero locus of the homogeneous partial derivatives is projective over $\mathbb{Z}_{(1009)}$, so a nonempty generic fiber would force a nonempty special fiber. A smooth hypersurface in $\mathbb{P}^3$ is geometrically irreducible, since two positive-degree components would intersect in singular points. The real smooth cone boundary is consequently Zariski dense in $\{f = 0\}$.

To identify the generic kernel, the certificate gives the point $(884, 370, 185, 278)$ modulo 1009. Its cubic base value is 721; the four exit denominators are $624, 500, 434, 609$. At those exits, the quadratic factors $p_{2,4}$ have values $822, 106, 124, 220$, and every other extra factor in (4.7) is nonzero. These conditions hold generically on the characteristic-zero hypersurface: divisibility by the primitive polynomial $f/4$ over $\mathbb{Q}$ would, by Gauss's lemma over $\mathbb{Z}_{(1009)}$, imply divisibility after reduction, contrary to this witness. They therefore hold generically on its real cone boundary, where the cubic base lies in its open cone by common-index deletion and the nonvanishing of its polynomial.

Each cubic exit is smooth, because its derivative in the last diagonal-sum direction is the nonzero quadratic factor. The determinant factorization (4.7) then has a simple zero there, so the cubic pencil has corank one: corank at least two would make its adjugate, and hence every first derivative of its determinant, vanish. At the generic real boundary points under consideration, we put $\theta_j = D_j/p > 0$. The maps $w \mapsto U_j^* w$ and $u \mapsto \theta_j^{-1} L^{-1} U_j u$ are inverse isomorphisms between $\ker(L - \theta_j^{-1} U_j U_j^*)$ and $\ker(U_j^* L^{-1} U_j - \theta_j I)$. Lemma 2.4 identifies θ_j as the maximal eigenvalue, which is therefore simple. The tensor sum of the matrices $U_j^* L^{-1} U_j$ consequently has a one-dimensional maximal eigenspace, so the uncompressed quartic pencil has corank one generically on its boundary. The cubic kernel vector (4.8) at each exit has nonzero last coordinate $p_{2,4}(Y_j)$, so $w_j \neq 0$ and $Lw_j = pU_j z_j$ forces $z_j \neq 0$. Thus the bottom block $p \bigotimes_j z_j$ is nonzero, and the quartic kernel is spanned by the displayed polynomial vector, or by R wherever $c \neq 0$.

The span of these real kernel lines is exactly the coefficient span of R, and so has dimension 244. Indeed, a constant linear functional vanishing on all these lines gives a polynomial vanishing on the dense boundary; it is divisible by f but has t-degree less than four, so it is zero.

Finally, if $L(y) \succeq 0$ and $Q^*L(y)Q$ is singular for a full-column-rank constant matrix Q, then $\operatorname{im} Q \cap \ker L(y) \neq \{0\}$: a zero quadratic form of a PSD matrix is a kernel vector. Every cone-preserving compression is singular at each generic boundary point and must therefore contain its unique kernel line. Its order is at least 244. The same conclusion holds for complex Q, since the complex span of these real kernel lines also has dimension 244. Conversely, the proof of Lemma 4.1 needs only $f(\mathbf{1}) > 0$ and $L(\mathbf{1}) \succ 0$ in place of its normalizations. It therefore applies to the fixed-matrix pencil, and compression to the coefficient span of R represents the cone in order 244. □

S2.1. **A different pencil of order at most twenty.** The preceding lower bound depends on the pencil being compressed. A different construction realizes the four bordering quotients with a common cubic denominator, which permits a boundary kernel of degree three.

Proposition S2.2. *For the matrix A and polynomial f in (S2.1), there are real symmetric PSD matrices $C_1, \dots, C_4$ of order at most 20 such that $\sum_j C_j \succ 0$ and $\Lambda_+(f, \mathbf{1}) = \{z : \sum_j z_j C_j \succeq 0\}$. Its positive definite locus is $\Lambda_{++}(f, \mathbf{1})$, and its determinant is divisible by f, with a cone-preserving multiplier. The coefficients can be chosen real algebraic.*

Proof. We write $B = A[\{1,2,3\}]$, $b = (1,2,3)^{\mathsf{T}}$, and $D = \operatorname{diag}(7,8,9)$, and put $B_0 = D^{-1/2}BD^{-1/2}$ and $b_0 = D^{-1/2}b$, so that the diagonal entries of B_0 are one. We define

$$s = z_1 + z_2 + z_3 + z_4, \quad x = z_1 + z_2 - z_3 - z_4,$$
$$y = z_1 - z_2 + z_3 - z_4, \quad w = z_1 - z_2 - z_3 + z_4.$$

With these coordinates we set

$$a = \frac{1969}{2016}, \qquad \beta = \frac{45}{1969}, \qquad \gamma = \frac{6}{1969}, \qquad p = s^3 - \beta s(x^2 + y^2 + w^2) + \gamma xyw.$$

Direct expansion of the order-three partition formula gives

$$\eta(z_1B_0, \dots, z_4B_0) = s^3 - \frac{2}{21} s \sum\nolimits_j z_j^2 + \frac{1}{126} \sum\nolimits_j z_j^3 = ap.$$

We define the derivatives $D_j(z) = D^{(j)}_{b_0 b_0^*} \eta(z_1B_0, \dots, z_4B_0)$ and put $F = f/504$. The bordering identity is

$$F = 10as\,p - \sum\nolimits_{j=1}^{4} z_j^2 D_j. \tag{S2.3}$$

The cubic partition formula gives

$$D_j = \frac{23}{14} s^2 - \frac{13}{252} \sum\nolimits_i z_i^2 - \frac{3}{7} s z_j + \frac{19}{252} z_j^2.$$

In particular,

$$D_1 = A_0 s^2 + A_1 s(x + y + w) + A_2(x^2 + y^2 + w^2) + A_3(xy + xw + yw), \tag{S2.4}$$

where the constants are

$$A_0 = \frac{2053}{1344}, \quad A_1 = -\frac{197}{2016}, \quad A_2 = -\frac{11}{1344}, \quad A_3 = \frac{19}{2016}.$$

For $j = 1, 2, 3, 4$, respectively, let $\epsilon_j = (1,1,1), (1,-1,-1), (-1,1,-1), (-1,-1,1)$. Then D_j is obtained from (S2.4) by the sign change $(x, y, w) \mapsto \epsilon_j \cdot (x, y, w)$. Each sign change fixes p.

To represent D_j/p as a quadratic form in an inverse, we use a cubic pencil whose determinant is exactly p. We let $r = \sqrt{\beta}$ and choose $\theta \in (0, \pi/2)$ with

$$\cos\theta = \frac{\gamma}{2\beta^{3/2}} = \frac{\sqrt{1969}}{45\sqrt{5}}.$$

We define the Hermitian matrices

$$H_1 = r\begin{pmatrix}1&0&0\\0&-1&0\\0&0&0\end{pmatrix}, \qquad H_2 = \frac{r}{\sqrt{2}}\begin{pmatrix}0&0&1\\0&0&1\\1&1&0\end{pmatrix}, \qquad H_3 = \frac{r}{\sqrt{2}}\begin{pmatrix}0&0&e^{i\theta}\\0&0&-e^{-i\theta}\\e^{-i\theta}&-e^{i\theta}&0\end{pmatrix}.$$

The cubic pencil is $M(s,x,y,w) = sI_3 + xH_1 + yH_2 + wH_3$, and expansion gives $\det M = p$. Let M_1, M_2, M_3 be the cyclic variants $M(s,x,y,w)$, $M(s,y,w,x)$, $M(s,w,x,y)$; all have determinant p and are positive definite at $z = \mathbf{1}$.

The adjugate identity requires a positive definite constant matrix. We set

$$S = H_1 + H_2 + H_3, \qquad R = H_1^2 + H_2^2 + H_3^2, \qquad J = S^2 - R,$$

and define

$$T = \lambda_0 I_3 + \lambda_1 S + \lambda_2 R + \lambda_3 J,$$

where the coefficients are

$$(\lambda_0,\lambda_1,\lambda_2,\lambda_3) = \left(-\frac{62719}{3523392}, \frac{331487057}{554934240}, \frac{5621495}{1370208}, \frac{42646571}{24663744}\right).$$

The three leading principal minors of T are

$$\frac{2053}{12096} + \frac{387893}{544320}\sqrt{\frac{45}{1969}}, \qquad \frac{33961555069}{3356242283520}, \qquad \frac{2766846449952493}{3425380874560512000}.$$

They are positive, so $T \succ 0$. Direct trace identities give

$$\operatorname{tr} T = A_0/3, \quad \operatorname{tr}(TS) = -A_1, \quad \operatorname{tr}(TR) = A_2 + \beta A_0, \quad \operatorname{tr}(TJ) = A_3.$$

For $H = xH_1 + yH_2 + wH_3$,

$$\operatorname{adj}(sI + H) = s^2 I - sH + H^2 - \beta(x^2+y^2+w^2)I,$$

so summing over the three cyclic variants proves

$$D_1 = \sum_{i=1}^{3} \operatorname{tr}(\operatorname{adj}(M_i)T). \tag{S2.5}$$

Applying the four sign changes gives the analogous identity for D_j; write the corresponding pencils as $M_{j,i}$.

We factor $T = \sum_{k=1}^{3} c_k c_k^*$ with algebraic column vectors c_k. For a complex matrix X, put

$$\mathcal{R}(X) = \begin{pmatrix}\operatorname{Re} X & -\operatorname{Im} X\\ \operatorname{Im} X & \operatorname{Re} X\end{pmatrix},$$

and write $\rho(c) = (\operatorname{Re} c, \operatorname{Im} c)^{\mathsf{T}}$. We form a real symmetric arrow pencil $\mathcal{L}(z)$ with the 36 diagonal blocks $\mathcal{R}(M_{j,i})$ for $1 \le j \le 4$ and $1 \le i,k \le 3$, a final scalar diagonal entry $10as$, and coupling from the (j,i,k) block to the scalar entry equal to $z_j\rho(c_k)$. Its order is 217. Where the cubic base is positive definite, its Schur complement is, by (S2.5),

$$10as - \sum_j z_j^2 \sum_i \operatorname{tr}(M_{j,i}^{-1}T) = 10as - \frac{\sum_j z_j^2 D_j}{p} = \frac{F}{p}.$$

In particular, $\mathcal{L}(\mathbf{1}) \succ 0$, since $p(\mathbf{1}) = 64$ and $F(\mathbf{1}) > 0$. To check the component, suppose a path from $\mathbf{1}$ inside $\{F > 0\}$ first reaches the boundary of the common positive definite locus of the $M_{j,i}$. There all these matrices are PSD and $p = 0$, so their adjugates are PSD. Equations (S2.3) and (S2.5) give $F \le 0$, a contradiction. Thus the Schur complement is positive throughout the open cone of f. Conversely, the positive definite locus of $\mathcal{L}$ is convex, contains $\mathbf{1}$, and has $F > 0$, so it lies in that open cone. Closure gives the closed-cone equality. The Schur complement also gives $\det \mathcal{L} = Fp^{71}$. For a further compression we define a real polynomial vector v by

$$v_{j,i,k} = -z_j\rho(\operatorname{adj}(M_{j,i})c_k), \qquad v_{\text{last}} = p.$$

The adjugate identities give $\mathcal{L}v = Fe_{\text{last}}$. Every coordinate of v is a real homogeneous cubic in four variables, so its coefficient span has dimension at most $\binom{6}{3} = 20$.

To retain the determinant factor after compression, we use coprimality of F and p. The cubic p is smooth over $\mathbb{C}$: its gradient equations have no nonzero solution when $\beta\gamma \neq 0$ and $\gamma^2 \neq 4\beta^3$, as here. Hence p is irreducible. At $(s, x, y, w) = (0, 1, 0, 0)$, one has $p = 0$ but $F = 11/5376$, so p does not divide F. We choose a real matrix Q whose columns form an orthonormal basis of the coefficient span and write $v = Q\widetilde{v}$. Multiplying

$$(Q^{\mathsf{T}}\mathcal{L}Q)\widetilde{v} = FQ^{\mathsf{T}}e_{\text{last}}$$

by the adjugate and then by $e_{\text{last}}^{\mathsf{T}}Q$ gives

$$p \det(Q^{\mathsf{T}}\mathcal{L}Q) = Fe_{\text{last}}^{\mathsf{T}}Q \operatorname{adj}(Q^{\mathsf{T}}\mathcal{L}Q)Q^{\mathsf{T}}e_{\text{last}}.$$

Coprimality therefore gives divisibility of the compressed determinant by F. The positive definite locus of the compressed pencil is convex, contains $\mathbf{1}$, and avoids $\{F = 0\}$; compression supplies the reverse inclusion from the open cone. The closed cones agree by adding a positive multiple of $\mathbf{1}$. The quotient determinant is hyperbolic and positive on the open cone, so its cone contains that of f.

The positive orthant lies in $\Lambda_+(f, \mathbf{1})$, so the coefficients $C_j = (Q^{\mathsf{T}}\mathcal{L}Q)(e_j)$ are PSD. All coefficients used above are algebraic, and algebraic orthonormalization preserves that property, which completes the proof. □

Remark S2.3 (The minimum over unrelated real symmetric pencils)**.** *Let $N_{\min}^{\text{real}}$ be the minimum order of a real symmetric pencil representing the cone for* (S2.1)*. Proposition S2.2 gives the upper bound in $5 \leq N_{\min}^{\text{real}} \leq 20$. After removing any common kernel, such a pencil is strictly feasible at an interior point. Its determinant vanishes on the cone boundary, so irreducibility makes it divisible by f and excludes orders below four. An order-four symmetric pencil would have determinant a nonzero constant times the smooth quartic f. For an affine pencil, its homogeneous degree-four part gives the same identity for the linear part. The projective matrix span would be a $\mathbb{P}^3$; a smaller span would make f a singular projective cone. The rank-at-most-two locus in $\mathbb{P}(\operatorname{Sym}_4 \mathbb{C})$ has codimension three and meets every $\mathbb{P}^3$. At such a matrix all first derivatives of the determinant vanish, contradicting smoothness. This argument concerns real symmetric pencils; it does not exclude an order-four Hermitian pencil.*

References